A pattern of anti-carbohydrate antibody responses present in patients with advanced atherosclerosis.

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We have previously shown that an antibody pool present in normal human serum binds cytokine receptors in vitro and may therefore interfere with assays that capture cytokines using their receptors. Here we show that this antibody pool is the same as the natural antibody termed anti-gal, that binds to the alpha-galactosyl carbohydrate epitope (alpha-gal) and which is the predominant obstacle to xenotransplantation. We report that there are high levels of IgD anti alpha-gal in most volunteers, in addition to the IgG2, IgA and IgM immunoglobulin isotypes against alpha-gal previously described. To determine if anti-gal may interfere with assays that depend on capture of cytokine with its receptor, we measured levels of several anti-carbohydrate antibodies in a cohort of patients with advanced atherosclerosis that had previously been used to measure levels of active TGF-beta using such an assay. For many isotype / carbohydrate combinations, there is a large and significant difference between the levels of anti-carbohydrate antibodies in patients with atherosclerosis and controls, after adjustment for age, sex and blood group. These results are similar to the previous data obtained for active TGF-beta, and therefore we cannot discount the possibility that anti-gal contributed to the previous data. Following further adjustment for several risk factors associated with cardiovascular disease, several anti-carbohydrate antibodies were still significantly different between patients and controls. Therefore, anti-carbohydrate antibodies may represent a new class of risk factors that may be associated with presence of advanced atherosclerosis, although larger studies will be required to confirm this hypothesis.